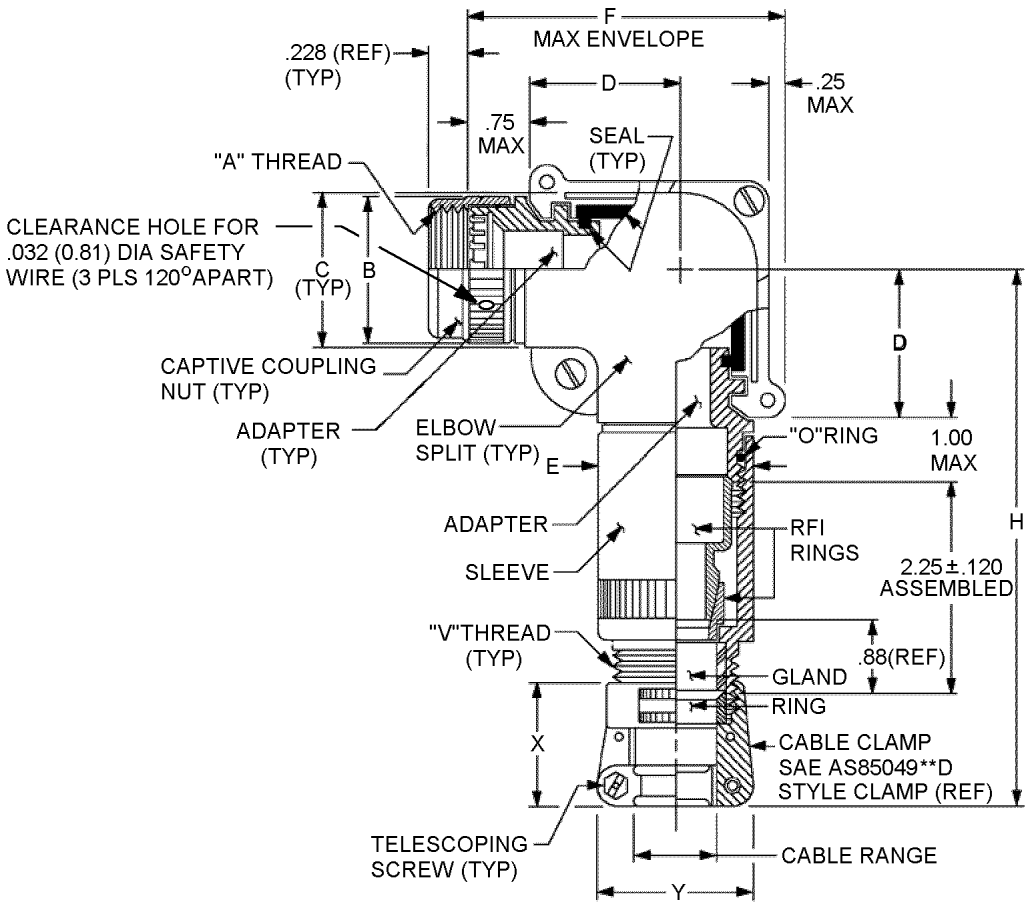
INCH-POUND
MIL-DTL-28840/8F
7 November 2013
SUPERSEDING
MIL-DTL-28840/8E
13 May 2009

DETAIL SPECIFICATION SHEET

CONNECTORS, ELECTRICAL, CIRCULAR, THREADED, HIGH SHOCK, HIGH DENSITY, SHIPBOARD, BACKSHELL, 90°, CABLE SEALING AND SHIELD TERMINATION

This specification is approved for use by all Departments and Agencies of the Department of Defense.

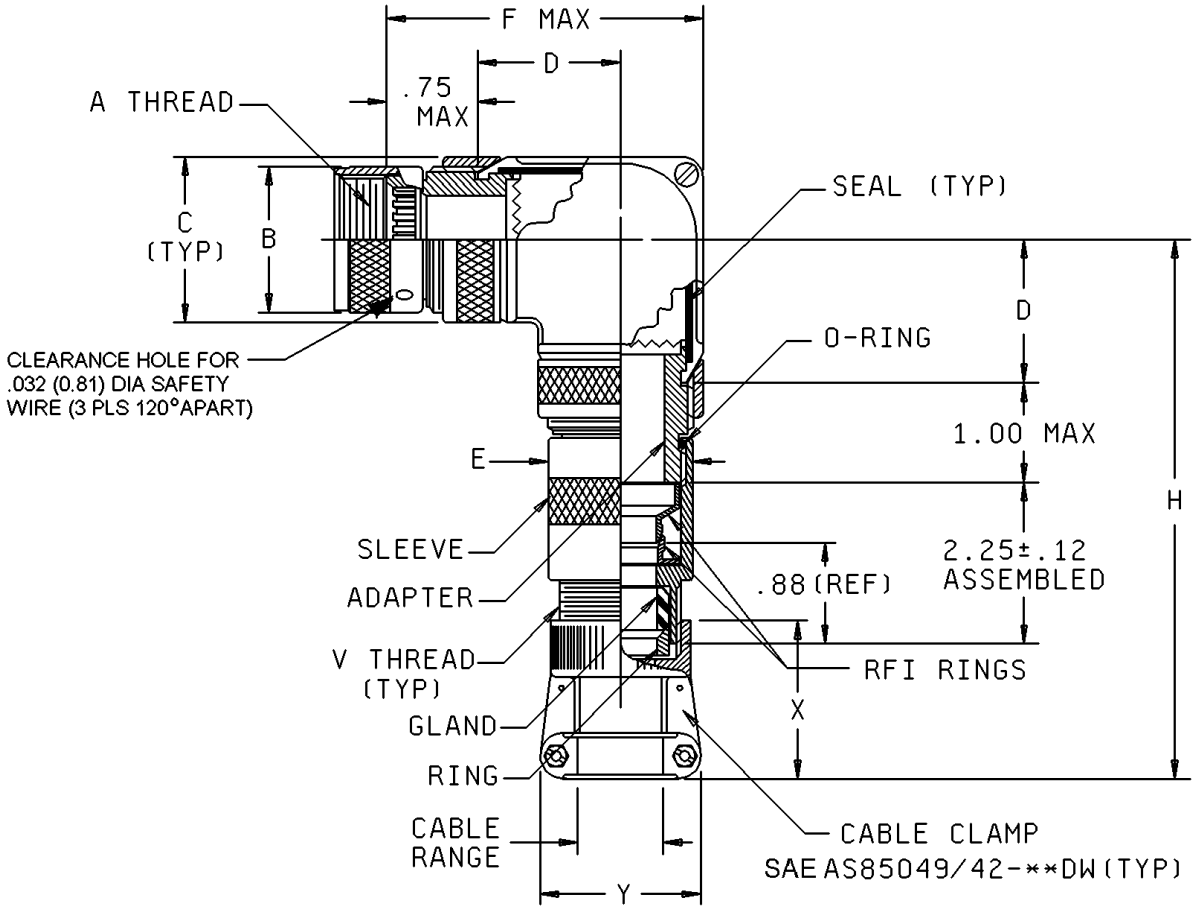
The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-28840.



STYLE A

FIGURE 1. Dimensions and configurations.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:34Z
Check the source to verify that this is the current version before use.

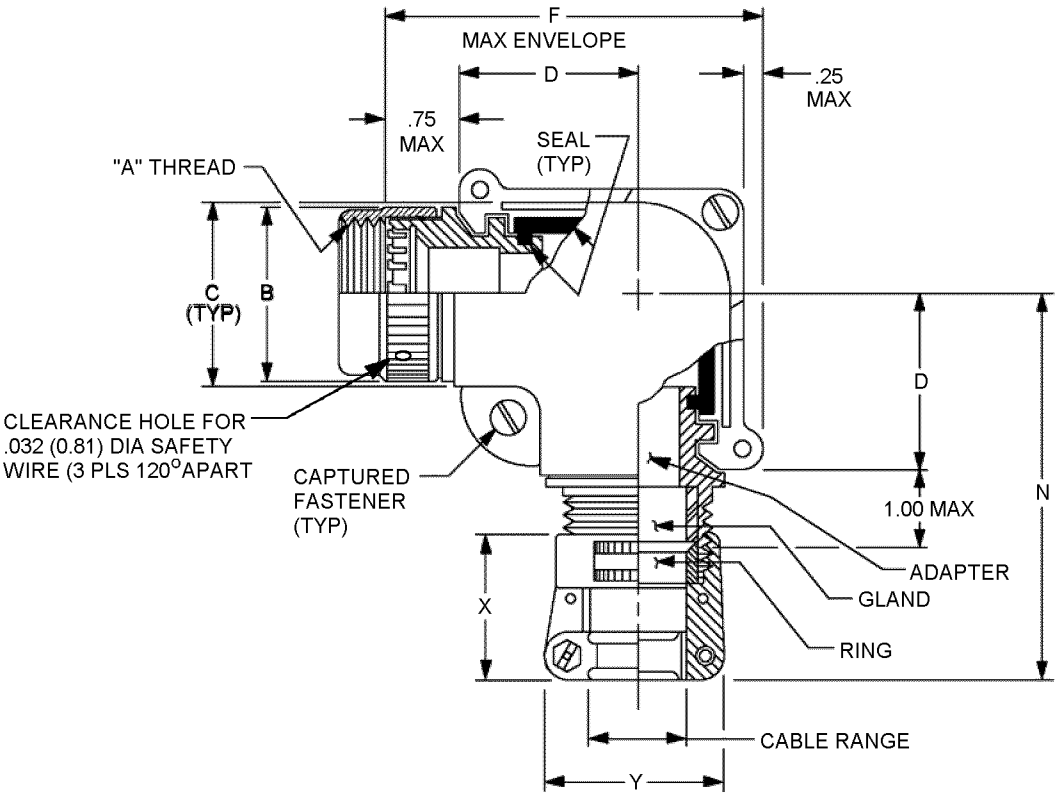


ALTERNATE STYLE A

Inches	mm
.12	3.05
.228	5.79
.25	6.35
.75	19.05
.88	22.35
2.25	57.15

FIGURE 1. Dimensions and configurations - Continued.

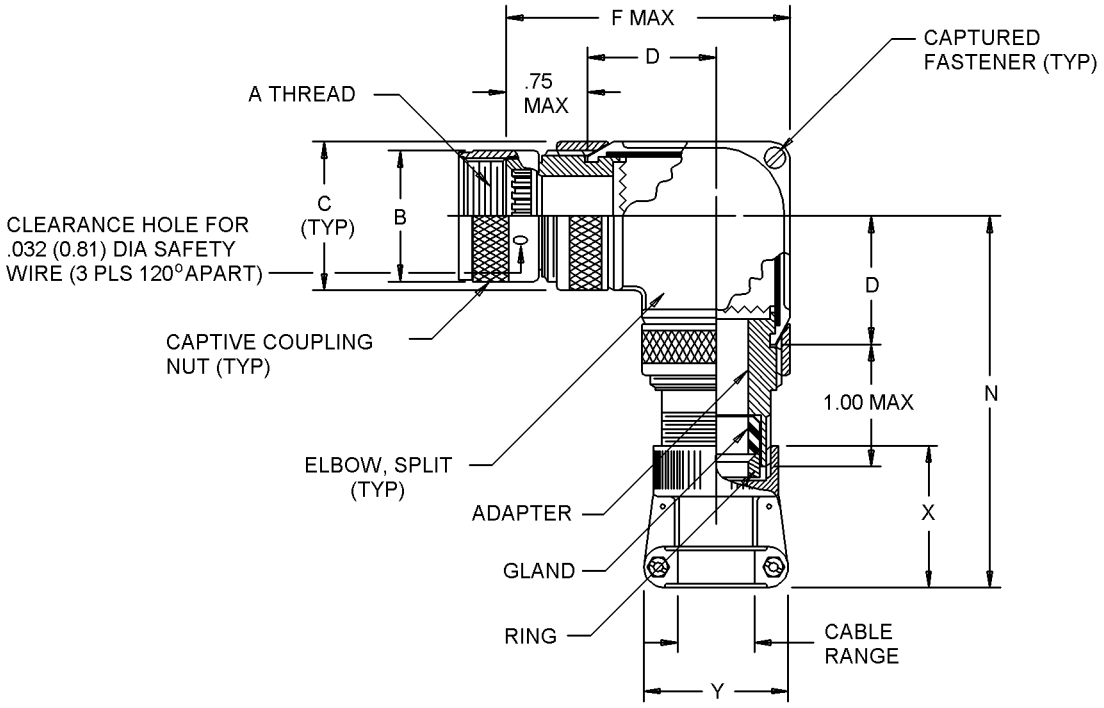
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:34Z
Check the source to verify that this is the current version before use.



STYLE B

FIGURE 1. Dimensions and configurations - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:34Z
Check the source to verify that this is the current version before use.



ALTERNATE STYLE B

FIGURE 1. Dimensions and configurations - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:34Z
Check the source to verify that this is the current version before use.

Backshell size (see note 3)	Dash no.	Shell size designator (see note 5)	A thread class 2B	B dia max	C dia max	D max dim
A	01	A (11)	.750-20 UNEF	1.028 (26.11)	1.188 (30.17)	1.031 (26.19)
A	02	B (13)	.875-20 UNEF	1.141 (28.98)		1.031 (26.19)
B	03	B (13)				
A	04	C (15)	1.00-20 UNEF	1.263 (32.08)	1.438 (36.52)	1.156 (29.36)
B	05	C (15)				
C	06	C (15)				
D	07	C (15)			1.688 (42.87)	1.281 (32.54)
A	08	D (17)	1.125-18 UNEF	1.387 (35.23)		
B	09	D (17)				
C	10	D (17)				
A	11	E (19)	1.250-18 UNEF	1.513 (38.43)		
B	12	E (19)			1.938 (49.22)	1.406 (35.71)
A	13	F (23)	1.438-18 UNEF	1.703 (43.26)		
B	14	F (23)				
C	15	F (23)				
A	16	G (25)	1.562-18 UNEF	1.825 (46.35)		
B	17	G (25)				
C	18	G (25)			2.188 (54.43)	1.531 (38.89)
D	19	G (25)				
A	20	H (29)	1.875-16 UN	2.143 (54.43)		
B	21	H (29)				

FIGURE 1. Dimensions and configurations - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:34Z
Check the source to verify that this is the current version before use.

Backshell size (see note 3)	Dash no.	Shell size designator (see note 5)	Cable range		E dia max	F max dim	V thread class 2* (see note 6)
			max	min			
A	01	A (11)	.312 (7.92)	.188 (4.77)	1.156 (29.36)	2.614 (66.40)	.625-24 UNEF
A	02	B (13)	.312 (7.92)	.188 (4.77)	1.156 (29.36)		.625-24 UNEF
B	03	B (13)	.438 (11.12)	.281 (7.14)			.750-20 UNEF
A	04	C (15)	.312 (7.92)	.188 (4.77)	1.281 (32.54)	2.864 (72.75)	.625-24 UNEF
B	05	C (15)	.438(11.12)	.281 (7.14)			.750-20 UNEF
C	06	C (15)	.562 (14.27)	.344 (8.74)			.875-20 UNEF
D	07	C (15)	.750 (19.05)	.438 (11.12)	1.406 (35.71)	3.114 (79.10)	1.188-18 UNEF
A	08	D (17)	.625 (15.87)	.375 (9.53)			1.000-20 UNEF
B	09	D (17)	.750 (19.05)	.438 (11.12)			1.188-18 UNEF
C	10	D (17)	.938 (23.82)	.625 (15.87)	1.531 (38.89)		1.438-18 UNEF
A	11	E (19)	.625 (15.87)	.375 (9.52)	1.406 (35.71)		1.000-20 UNEF
B	12	E (19)	.938 (23.82)	.625 (15.87)	1.531 (38.89)	3.364 (85.45)	1.438-18 UNEF
A	13	F (23)	.750 (19.05)	.438 (11.12)	1.406 (25.71)		1.188-18 UNEF
B	14	F (23)	.938 (23.82)	.625 (15.87)	1.531 (38.89)		1.438-18 UNEF
C	15	F (23)	1.188 (30.17)	.812(20.62)	1.781 (45.24)		1.750-18 UNS
A	16	G (25)	.938 (23.82)	.625 (15.87)	1.531 (38.89)		1.438-18 UNEF
B	17	G (25)	1.188 (30.17)	.812 (20.62)	1.781 (45.24)		1.750-18 UNS
C	18	G (25)	1.250 (31.75)	.875 (22.23)		3.614 (91.80)	1.750-18 UNS
D	19	G (25)	1.380 (35.05)	1.000 (25.40)	2.156 (54.76)		2.000-16 UNS
A	20	H (29)	1.188 (30.17)	.812 (20.62)	1.781 (45.24)		1.750-18 UNS
B	21	H (29)	1.380 (35.05)	1.000 (25.40)	2.156 (54.76)		2.000-18 UNS

FIGURE 1. Dimensions and configurations - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:34Z
Check the source to verify that this is the current version before use.

Backshell size (see note 3)	Dash no.	Shell size designator (see note 5)	X ref dim	Y ref dim	H max style A	N max style B
A	01	A (11)	1.281 (32.54)	.938 (23.82)	5.332 (135.43)	2.962 (75.23)
A	02	B (13)	1.281 (32.54)	.938 (23.82)	5.332 (135.43)	2.962 (75.23)
B	03	B (13)	1.281 (32.54)	1.125 (28.57)	5.332 (135.43)	2.962 (75.23)
A	04	C (15)	1.281 (32.54)	.938 (23.82)	5.457 (138.61)	3.087 (78.41)
B	05	C (15)	1.281 (32.54)	1.125 (28.57)	5.457 (138.61)	3.087 (78.41)
C	06	C (15)	1.281 (32.54)	1.312 (33.32)	5.457 (138.61)	3.087 (78.41)
D	07	C (15)	1.281 (32.54)	1.593 (40.46)	5.682 (144.32)	3.312 (84.12)
A	08	D (17)	1.281 (32.54)	1.312 (33.32)	5.682 (144.32)	3.312 (84.12)
B	09	D (17)	1.312 (33.32)	1.593 (40.46)	5.713 (145.11)	3.343 (84.91)
C	10	D (17)	1.406 (35.71)	1.750 (44.45)	5.807 (147.50)	3.437 (87.30)
A	11	E (19)	1.281 (32.54)	1.312 (33.32)	5.682 (144.32)	3.312 (84.12)
B	12	E (19)	1.406 (35.71)	1.750 (44.45)	5.832 (148.13)	3.462 (87.93)
A	13	F (23)	1.312 (33.32)	1.593 (40.46)	5.738 (145.75)	3.368 (85.55)
B	14	F (23)	1.406 (35.71)	1.750 (44.45)	5.832 (148.13)	3.462 (87.93)
C	15	F (23)	1.593 (40.46)	2.093 (53.16)	6.019 (152.88)	3.649 (92.68)
A	16	G (25)	1.406 (35.71)	1.750 (44.45)	5.832 (148.13)	3.462 (87.93)
B	17	G (25)	1.593 (40.46)	2.093 (53.16)	6.019 (152.88)	3.649 (92.68)
C	18	G (25)	1.593 (40.46)	2.093 (53.16)	6.144 (156.06)	3.774 (95.86)
D	19	G (25)	1.625 (41.28)	2.343 (59.51)	6.176 (156.87)	3.806 (96.62)
A	20	H (29)	1.593 (40.46)	2.093 (53.16)	6.144 (156.06)	3.774 (95.86)
B	21	H (29)	1.625 (41.28)	2.343 (59.51)	6.176 (156.87)	3.806 (96.62)

FIGURE 1. Dimensions and configurations - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:34Z
Check the source to verify that this is the current version before use.

Backshell size (see note 3)	Dash no.	Shell size designator (see note 5)	A thread class 2B	B dia max	C dia max	D dia max
A	22	J (33)	2.062-16 UNS	2.329 (59.16)	2.688 (68.26)	1.781 (45.24)
A	23	J (33)				
B	24	A (11)	.750-20 UNEF	1.028 (26.11)	1.188 (30.17)	1.031 (26.19)
C	25	A (11)			1.438 (36.52)	1.156 (29.36)
C	26	B (13)	.875-20 UNEF	1.141 (28.98)		
D	27	B (13)			1.688 (42.87)	1.281 (32.54)
E	28	C (15)	1.000-20 UNEF	1.263 (32.08)		

Backshell size (see note 3)	Dash no.	Shell size designator (see note 5)	Cable range		E max dim	F max dim	V thread class 2* (see note 6)
			max	min			
A	22	J (32)	1.380 (35.05)	1.000 (25.40)	2.281 (57.94)	4.114 (104.50)	2.000-18 UNS
A	23	J (33)	1.625 (41.26)	1.250 (31.75)			2.250-16 UNS
B	24	A (11)	.438 (11.12)	.281 (7.14)	1.156 (29.36)	2.614 (66.40)	.750-20 UNEF
C	25	A (11)	.562 (14.27)	.344 (8.74)	1.281 (32.54)	2.864 (72.75)	.875-20 UNEF
C	26	B (13)	.562 (14.27)	.344 (8.74)	1.281 (32.54)		.875-20 UNEF
D	27	B (13)	.625 (15.87)	.375 (9.52)	1.406 (35.71)	3.114 (79.10)	1.000-20 UNEF
E	28	C (15)	.625 (15.87)	.375 (9.50)			1.000-20 UNEF

FIGURE 1. Dimensions and configurations - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:34Z
Check the source to verify that this is the current version before use.

Backshell size (see note 3)	Dash no.	Shell size designator (see note 5)	X ref dim	Y ref dim	H max style A	N max style B
A	22	J (33)	1.625 (41.28)	2.343 (59.51)	6.426 (163.22)	4.056 (103.02)
B	23	J (33)	1.900 (48.26)	2.750 (69.85)	6.701 (170.21)	4.331 (110.01)
B	24	A (11)	1.281 (32.54)	1.125 (28.57)	5.332 (135.43)	2.962 (75.23)
C	25	A (11)	1.281 (32.54)	1.312 (33.32)	5.457 (138.61)	3.087 (78.41)
C	26	B (13)	1.281 (32.54)	1.312 (33.32)	5.457 (138.61)	3.087 (78.41)
D	27	B (13)	1.281 (32.54)	1.312 (33.32)	5.682 (144.32)	3.312 (84.12)
E	28	C (15)	1.281 (32.54)	1.312 (33.32)	5.682 (144.32)	3.312 (84.12)

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Designator of backshell size within each shell size.
4. Dimensions apply after plating.
5. Shell sizes are provided within parentheses for information and are not a part of the designator.
6. See MIL-DTL-28840 for V thread, class 2*.

FIGURE 1. Dimensions and configurations - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:34Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

Dimensions and configurations: See figure 1.

Interface dimensions shall conform to MIL-DTL-28840.

Material and finish: See table I.

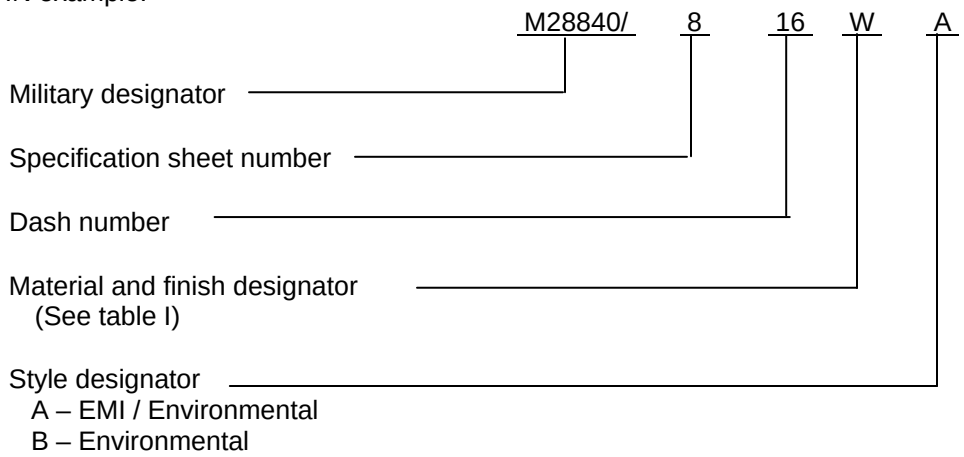
TABLE I. Material and finish.

Material and finish designator	Material	Finish
W	Aluminum alloy in accordance with ASTM B85/B85M, ASTM B26/B26M, ASTM B108/B108M, ASTM B209, ASTM B211, or ASTM B221	Cadmium plate, olive drab over electroless nickel
D	Corrosion resistant steel in accordance with ASTM A276, ASTM A473, or ASTM A580/A580M, type 316	Cadmium plate, black 1/
F	Aluminum alloy in accordance with ASTM B85/B85M, ASTM B26/B26M, ASTM B108/B108M, ASTM B209, ASTM B211, or ASTM B221	Pure electro-deposited aluminum, non-reflective
L		Nickel fluorocarbon polymer, non-reflective
S		Zinc nickel, non-reflective
N	Corrosion resistant steel in accordance with ASTM A276, ASTM A473, or ASTM A580/A580M, type 316	Nickel fluorocarbon polymer, non-reflective
R		Zinc nickel, non-reflective

1/ Screws and washers shall be passivated in accordance with SAE-AMS2700, type 2.

Marking: The backshell shall be identified with the manufacturer's name or trademark and the Part or Identifying Number (PIN).

PIN example:



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:34Z
Check the source to verify that this is the current version before use.

QUALIFICATION: Two samples in each shell size for which qualification is desired and shall be tested in accordance with MIL-DTL-28840.

RESTRICTIONS:

Material and finish designators W and D are inactive for NAVSEA new design, unless approved by NAVSEA.

Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Referenced documents. In addition to MIL-DTL-28840, this document references the following:

ASTM A276
ASTM A473
ASTM A580/A580M
ASTM B26/B26M
ASTM B85/B85M
ASTM B108/B108M
ASTM B209
ASTM B211
ASTM B221
SAE-AMS2700

CONCLUDING MATERIAL

Custodians:
Army - AT
Navy - EC
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5935-2013-106)

Review activities:
Army – AV, MI
Air Force – 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:34Z
Check the source to verify that this is the current version before use.